Tim Book

Random Variables and Discrete Distributions

GENERAL ASSEMBLY

Agenda

- What is a **random variable**?
- What is a **probability distribution**?
  - ... and their associated calculations
- Discrete distribution flashcards!
- Playing around in Python

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Random Variables

What is a random variable?

You can think of a **random variable** as a “source” of randomness from which an experiment can be performed and from which we can get actual outcomes (**realizations**).

What is a random variable? An example:



Consider rolling a fair 6-sided die.

- The outcome of this event is a **random variable**.
- So, we roll the die!
- We rolled a **5**. This **outcome** is a **realization** from our random variable.

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Aside: Discrete vs Continuous



What is the difference between a **discrete variable** and a **continuous** one?



Is the outcome of rolling a fair 6-sided die discrete or continuous?

Aside: Discrete vs Continuous

- A discrete variable is **countable**. That is, you take two adjacent values and know that there is no legal value between them.
  - The **sample space** of rolling a d6 is {1, 2, 3, 4, 5, 6}. 3.5 is not a legal value!

- A continuous variable is infinitely **subdividable**. Outcomes can take any value in a given interval.
  - Your height is continuous. You can be 5’8, but you can also be 5’8.00001

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Aside: Discrete vs Continuous

A common ***MISCONCEPTION*** is that discrete random variables are **finite** and that continuous ones are **bounded**.

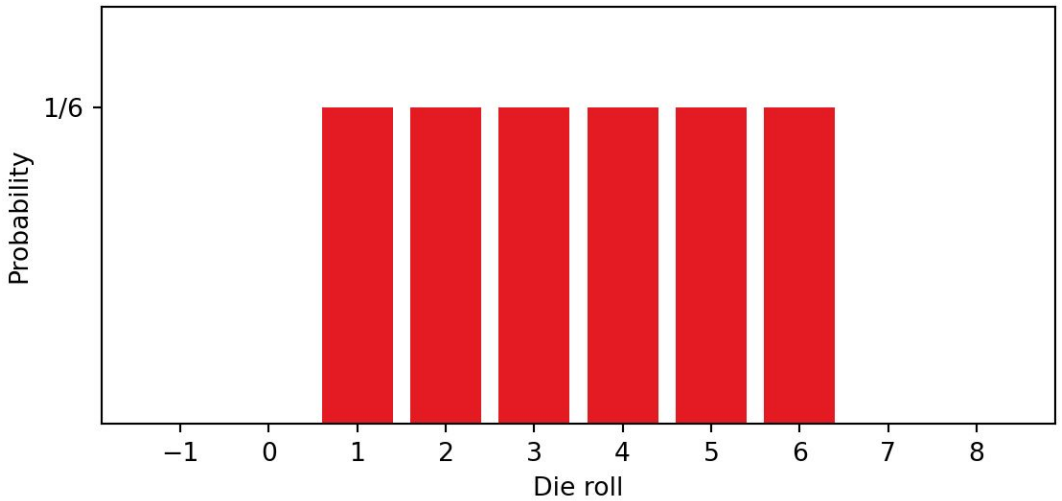
- The number of candies in a bag is discrete but there is no hard upper limit on them. So the sample space is: {0, 1, 2, 3, ...} (this may seem ridiculous and wrong but I'll convince you this is true by the end of this lesson.)
- The wind direction in weather forecasts is continuous but bounded between 0 and 360 degrees: [0, 360)

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Discrete Random Variables

Our first distribution!

A random variable that gives every outcome equal probability is said to follow a **(discrete) uniform distribution**.



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Notation

Let's return to the situation of rolling a fair 6-sided die. Let X denote the random variable associated with this situation. We use the following notation to denote that: ***X follows a discrete uniform distribution from 1 to 6.***

$$X \sim U(1, 6)$$

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All RVs have a quick abbreviation we use like this. Discrete uniform is U(*a*, *b*) or sometimes U[*a*, *b*].

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Notation

While we use capital letters to denote random variables, we often use the corresponding lowercase letter to represent a realization. That is, if

$$X \sim U(1, 6)$$

then after we roll a die, maybe we get

$$x = 5$$

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Notation

If random variables are... *random*... then we can calculate probabilities! Again, let X be the outcome of rolling a fair 6-sided die.



What is $P(X = 5)$?

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If random variables are... *random*... then we can calculate probabilities! Again, let X be the outcome of rolling a fair 6-sided die.



What is $P(X = 5)$?



What is $P(X \leq 5)$?

pmfs and CDFs

Probability Mass Functions (pmfs)

We typically write the formula for given probabilities out as a function called a **probability mass function (pmf)**. They're typically denoted with a little *f*. So for example, if X ~ U(1, 6), then

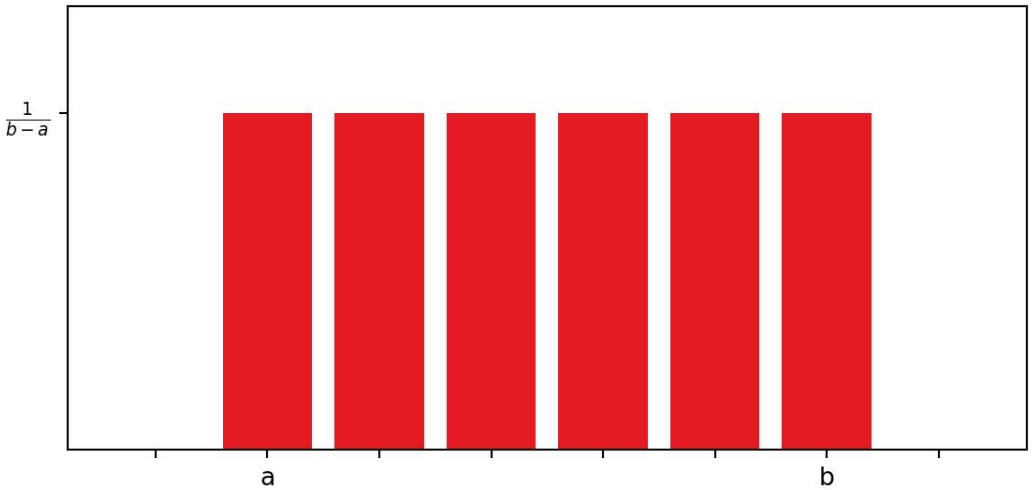
$$P(X = x) = f(x) = \frac{1}{6}$$

for *x* between 1 and 6.

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Probability Mass Functions (pmfs)

They're called "mass functions" because they describe where the "mass" (aka "probability") of a phenomenon lies. pmfs are often graphed in a histogram:



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Probability Mass Functions (pmfs)

Because pmfs are just probability rules, there are a few key facts about them worth mentioning:

- $f(x) = 0$ for any x outside the random variable's **support**.
- $0 \leq f(x) \leq 1$ for all x
- $\sum f(x) = 1$ for all x

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Probability Mass Functions (pmfs)

Because pmfs are just probability rules, there are a few key facts about them worth mentioning:

- $f(x) = 0$ for any x outside the random variable's **support**.
  - Impossible things have zero probability
- $0 \leq f(x) \leq 1$ for all x
  - Probabilities can only be between 0 and 1
- $\sum f(x) = 1$ for all x
  - The sum of all probabilities has to be 100%

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Cumulative Density Functions (CDFs)

We also define the concept of a **cumulative density function (CDF)** as the *cumulative* probability of a value. We typically use a big *F* for CDFs. For example, for our running die-roll example:

$$P(X \leq x) = F(x) = \frac{x}{6}$$

For all *x* between 1 and 6.

Note that

- $F(x) = 0$ for all x less than 1
- $F(x) = 1$ for all x greater than 6

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Cumulative Density Functions (CDFs)

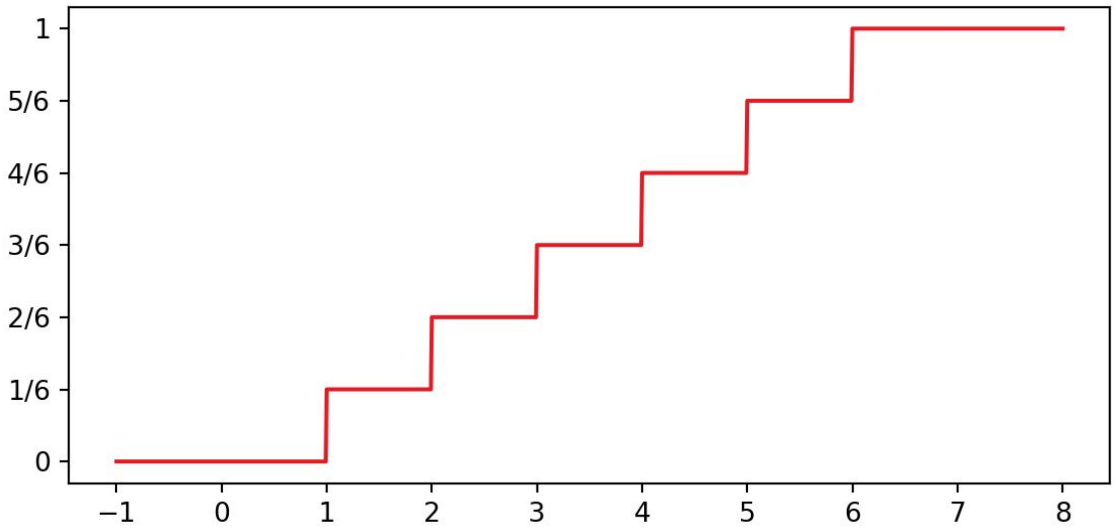
Most CDFs don't have **closed form formulas**. The uniform distribution is an exception to this rule because it's so simple. Another useful way to think about CDFs is as follows:

$$
\begin{aligned}
F(x) &= P(X \leq x) \\
&= P(X = 1) + P(X = 2) + \cdots + P(X = x) \\
&= \sum_{k=1}^{x} P(X = k)
\end{aligned}
$$

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Graph of a CDF

Here's our die-roll example's CDF graphed as a line plot:



Additional CDF facts

Like pmfs, CDFs have a few math facts that can help us understand them a little better:

- $F(x)$ is an increasing function in x
- $F(x) = 1$ for large enough x
- $P(X \geq x) = 1 - F(x)$

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Additional CDF facts

Like pmfs, CDFs have a few math facts that can help us understand them a little better:

- $F(x)$ is an increasing function in x
  - CDFs get larger the larger x is since they contain the same probability as the smaller x.
- $F(x) = 1$ for large enough x
  - Eventually, everything is smaller than x
- $P(X > x) = 1 - F(x)$
  - Finding the probability of being greater than a number is just the complement of the CDF (note the use of $>$ instead of $\geq$ here)

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Some interesting calculations about RVs

The mean of a random variable

Random variables are also said of have **means**. For our simple die-roll example, we can intuit that the average die roll is 3.5 (the average of 1 and 6). That is, if you rolled a die a zillion times and took the average result, you'll get 3.5.

Note that the average value doesn't need to be an integer, or even be in the support at all!

But how is the calculate for discrete random variables in general?

The mean of a random variable

$$E[X] = \sum x f(x)$$

The *E* stands for “expectation”. A commonly used synonym for the mean is **the expected value**.

This is actually a good habit to get into. In your head, every time you hear “mean” you can also think “expected value” and vice versa.

Mean examples

- On a given farm, a chicken has a 30% chance of laying zero eggs, and 50% chance of laying one egg, and a 20% chance of laying two eggs. What is the expected number of eggs a chicken will lay?



- You play a lottery game that costs $2. There's a 99% chance you win nothing, but a 1% chance you win $100. Is the game worth it?



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Mean examples

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  - E[X] = (0.3)(0) + (0.5)(1) + (0.2)(2) = **0.9**
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- You play a lottery game that costs $2. There's a 99% chance you win nothing, but a 1% chance you win $100. Is the game worth it?
  - E[X] = (0.99)(0) + (0.01)(100) = **1**
  - You expect to win $1, but paid $2 to enter. ***Bad deal!***

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The variance of a random variable

Variances also have a formula, but it's much more complicated. It's also less intuitive so we won't be doing any examples:

$$\text{Var}[X] = E[(X - \mu)^2] = E[X^2] - E[X]^2$$

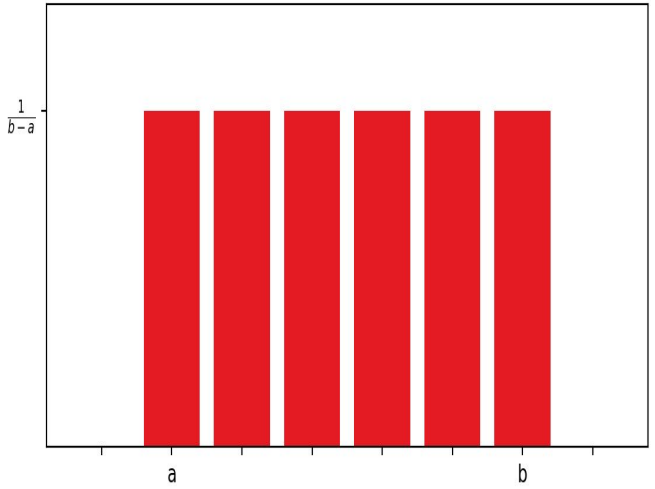
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Discrete Distribution Flashcards!

The Discrete Uniform Distribution

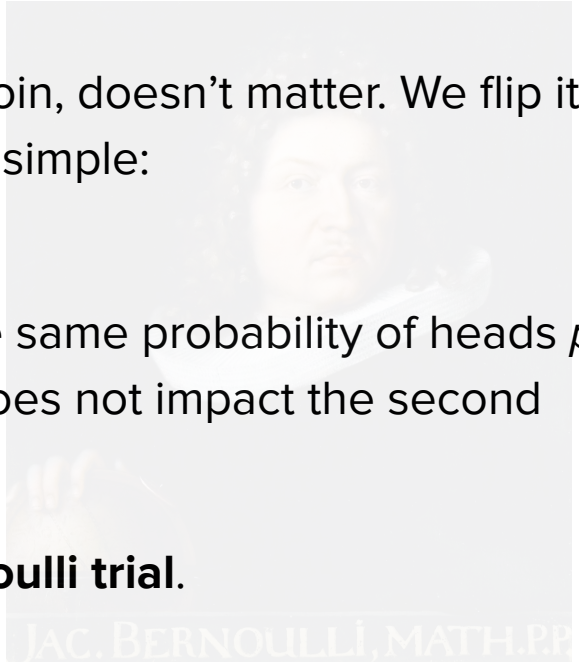
pmf	$f(x) = \frac{1}{b-a}$
support	$a \leq x \leq b$
mean	$\frac{a+b}{2}$
variance	$\frac{(b-a)^2}{12}$

$$X \sim U(a, b)$$



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The Bernoulli Distribution



Consider the simple coin flip. Maybe it's an unfair coin, doesn't matter. We flip it, and many things are true which make this situation simple:

- There are only two outcomes: heads or tails
- No matter how many times you flip it, it has the same probability of heads *p*.
- The coin flips are independent. Your first flip does not impact the second flip.

Any situation that fits these criteria is called a **Bernoulli trial**.

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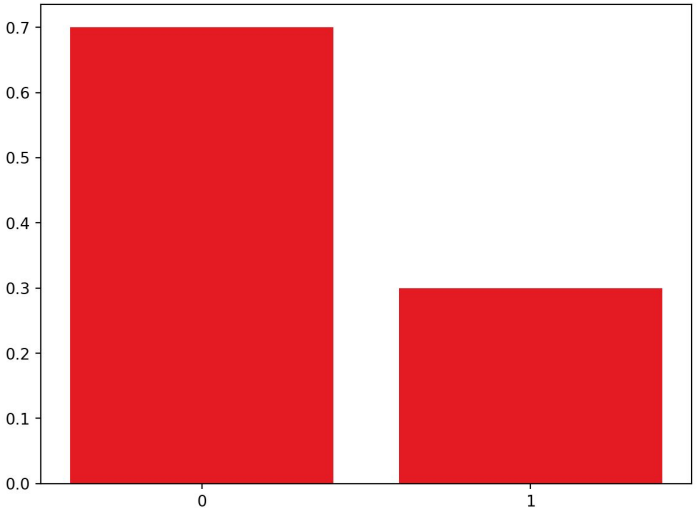


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The Bernoulli Distribution

pmf	$f(x) = p^x(1-p)^{1-x}$
support	$x \in \{0, 1\}$
mean	p
variance	$p(1-p)$

$$X \sim \text{Ber}(p)$$



This graph depicts Ber(0.3)

The Binomial Distribution

(This is a big, common, important one!)

The **binomial distribution** is the sum of successive bernoulli trials. Consider the following question:

How many outcomes are there if we flip two coins?

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4: HH, HT, TH, TT

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4: HH, HT, TH, TT

Then what is the probability of getting exactly one heads?

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What is the probability we get exactly two heads?

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The Binomial Distribution

Let’s say we flip *n* coins. How many ways are there to get *k* heads?

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The Binomial Distribution

Let's say we flip *n* coins. How many ways are there to get *k* heads?

Answer: "**n choose k**"

$$\binom{n}{k} = {}_nC_k = \frac{n!}{k!(n-k)!}$$

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The Binomial Distribution

Let's say we flip *n* coins. The probability of flipping a heads is *p*. What is the probability of getting *k* heads in these *n* flips?

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$$\binom{n}{k} p^k (1-p)^{n-k}$$



This many ways to get *k* heads when flipping *n* coins.

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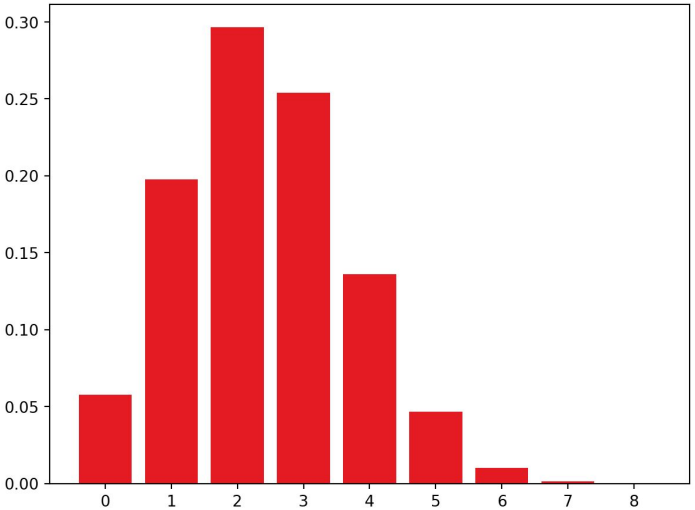
For each of these ${}_nC_k$ ways, this is the probability of getting *k* heads and *n - k* tails.

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The Binomial Distribution

pmf	$f(x) = \binom{n}{k} p^x (1-p)^{n-x}$
support	$0 \leq x \leq n$
mean	np
variance	$np(1-p)$

$$X \sim \text{Bin}(n, p)$$



This graph depicts Bin(8, 0.3)

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The Poisson Distribution

The Poisson distribution is the most common one you've never heard of! It's generally used to model **events with small counts in a fixed interval (of time).** It has one parameter, λ, which is its mean. Some examples:

- You run a bookstore and, on average, see 10 people enter your store per hour. You may model this with a Poi(10) distribution.
- You sell doors door-to-door and make, on average, 4 sales a day.
- The average American household has 2.1 children.
- During the Franco-Prussian war, an average of 5 soldiers were killed per year by being kicked in the head by their horses ***(this was a real use-case for the Poisson distribution soon after it was invented!)***

The Poisson Distribution

The Poisson distribution was used to model the frequency of **rare** events. During the Franco-Prussian war, many soldiers stood near horses (*n* large). Horse kicks were extremely rare events (*p* small), but happened anyway because of the frequency of trials.



https://i.ytimg.com/vi/8SsyeDqW_qE/maxresdefault.jpg

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Modeling: You know what they say about assumptions...

Up until now, we've said that certain phenomena **follow** distributions. There are rules for when something follows a uniform, Bernoulli, or binomial distribution. While this is generally true for Poisson, it's more often that we **assume** a phenomenon follows the Poisson distribution because it makes sense for it to do so.

I want you to link the idea of Poisson to **"small counts"**. This will be useful for our GLM lesson later.

Modeling: You know what they say about assumptions...

Maybe we consider the number of ads the typical American watches in a given day to be Poisson(8) distributed. Note that the Poisson distribution is **unbounded**. So, it gives nonzero probability to all numbers {0, 1, 2, ...}. So, technically, the probability you watch *one thousand* ads in a day has a positive probability! Remember, **models are usually approximations!**

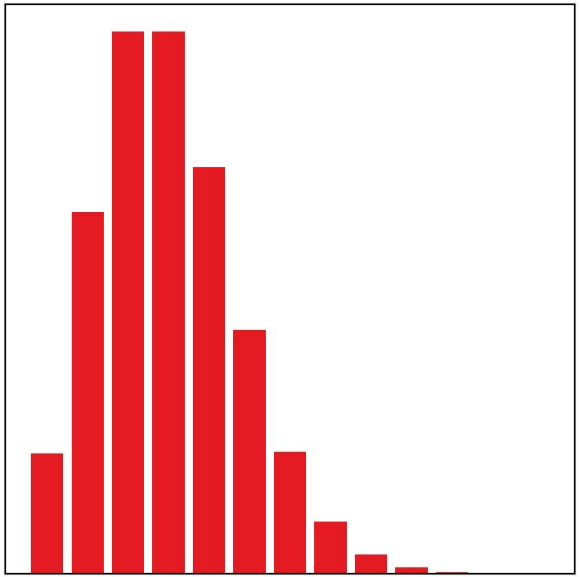
Anyway, if $X \sim$ Poisson(8), then P(1000) is 0 to over 100 decimal points...

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The Poisson Distribution

pmf	$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}$
support	$x \geq 0$
mean	λ
variance	λ

$$X \sim \text{Poi}(\lambda)$$



Python Time!



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But wait, there's more!

Bonus Distributions

(for those inclined)

(you won't see them again in our class)

The Geometric Distribution

Suppose you have a repeated Bernoulli trial event with probability of success *p*. The **geometric distribution** describes how many trials you have until your first success.

pdf	$f(x) = (1-p)^{x-1}p$
support	$x \geq 1$
mean	$\frac{1}{p}$
variance	$\frac{1-p}{p^2}$

Example: If ⅕ candies in a pack are green, the number of candies you eat until you randomly eat a green one might be Geometric(⅕).

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The Hypergeometric Distribution

Suppose you sample *without replacement* n objects from a population of size N. Of these N, K is the number of "successes" in the population. When drawing your n samples, the number of successes observed follows a **hypergeometric distribution**.

The Hypergeometric Distribution

The formulas are messy and not helpful, so we'll skip the table here. These things are the important values to know

- N = population size
- K = successes in population
- n = sample size

Example: In a game of poker, you draw 5 cards. The number of aces in your hand would follow Hypergeom(N = 52, K = 4, n = 5).

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The Multinomial Distribution

Consider a special Bernoulli trial in which there were more than two possible outcomes. Because we're not dealing with just 0s and 1s anymore, we have our first **random vector**!

Example: Suppose it's known the American population's voting registry is 40% Democrat, 40% Republican, and 20% Independent/Other. If we go out and sample 100 people, the distribution of our results might be:

$$\mathbf{X} \sim \text{Multinomial}\left(100, \begin{bmatrix} 0.4 \\ 0.4 \\ 0.2 \end{bmatrix}\right)$$